

Eurofins	Document Name: DevOpsEngineerTake-Home.pdf	Document Owner: FARS
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DevOps Engineer Take-Home Assignment

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Objective

The objective of the exercise is to setup a small IIS application pipeline, deployment and a small module to log its status. The implementation doesn't have necessary to be too precise / complete.

The exercise should be achieved in approximatively 4 hours, completion is not mandatory, and the candidate can come with a partial solution.

Step 5 and 6 are optionnal, but if you have time, and are familiar with docker, or want to learn it, you can give a try.

Brief

Our Product Owner Team want a Spike of a new customer management application. You are asked to create this spike application where we can enter new customers details and then display them on a grid.

Technical Stack

Use the followings:

- **Backend Language:** C#
- **Backend Software Framework:** One of the latest .NET, at least .NET CORE 3.1.
- **Scripting Language:** Powershell (prefered), Python,...

Description

Step 1 : Build

Create or use an existing simple .NET Core web application, such as a HelloWorld. Store the application in a version control system (e.g., Git, hosted on GitHub or Azure DevOps), and set up a continuous integration (CI) pipeline using a CI/CD tool of your choice (e.g., GitHub Actions, Azure DevOps Pipelines, GitLab CI, Jenkins, etc.).

The pipeline should automatically rebuild the package whenever changes are committed to the repository.

Ensure that you apply best practices for version control, code quality, and pipeline efficiency throughout the assignment.

Stages of Completion:

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1. Hello World web application in version control
2. CI pipeline that triggers automatically upon commit
3. A zip package of the "HelloWorld" application is created.
4. *Implementation of best practices for the repository and pipeline.*

Comment: If tool and platform are inconvenient, please do best effort to explain and describe with necessary code to support.

Step 2 : IIS Deployment

Please use one of the scripting languages you are familiar with (Powershell, Python,...) to create a script to deploy the web application done in Step 1 as the following:

- Create a web site in IIS and add an https binding
- Create a local group and add a specified user as a group member
- Create an application pool in IIS and run it with the specified user
- Modify the path of the web site log file
- Create an application at the lower level of the web site and bind the application pool created

Stage of Completion:

- "HelloWorld" is reachable via http using a localhost url.

Step 3 : Log Status

Using the programming language of your choice, create a small windows service that :

- Will check the IIS website status every 60 seconds
- Will put the result (HTTP Code and message) in a text / log file (in the same folder than executable)
- If the retur is different than 200 (OK), the service stop

Stage of Completion:

- Windows service log http status of "HelloWorld" every 60 seconds - when run in EDI
- Windows service will stop in http status <> 200 OK - when run in EDI

Step 4 : Windows Service Deployment

Please use one of the scripting languages you are familiar with (Powershell, Python,...) to create a script to deploy a windows service as the following:

- Create a windows service and point to that executable file (.exe)
- Run the windows service with a specific user
- Set the windows service startup type to auto-start
- Set the windows service error recovery to automatically recover every 300 seconds

Stage of Completion:

- Windows service is installed and start
- Windows service log http status of "HelloWorld" every 60 seconds
- Windows service will stop in http status <> 200 OK
- Windows service restart after 300 seconds

Step 5 : Docker Image (optional)

Update pipeline done in step 1, to create also a docker image of the "Hello World" application (in addition of the zip package).

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Stage of Completion:

- Elements of completion of step 1 are still there.
- Docker image of the application is created
- Docker image is uploaded on Docker Hub or alternative

Step 6 : Docker Deployment (optional)

Create a script that deploy the HelloWorld Docker image created in previous step, on the docker engine (on Windows or Linux server).

Note: if you didn't publish your image you can use alternatively https://hub.docker.com/_/hello-world

Stage of Completion:

- Elements of completion of step 1 are still there.

Deliverables:

Deliver the solution:

- EDI: editable with VS Code and/or Visual Studio for both "HelloWorld" and Windows Service
- Pipeline for HelloWorld solution, with creation of zip package and docker image
- Deployment Scripts (for both IIS and Docker Engine)
- Any documentation/explanation that you think is relevant
- Source: Packed as zip file
- Channel: Returned email

Keep all element of the solution accessibles, if you are selected for the next steps, demonstration and questions will be asked about it during the Technical Interview.